

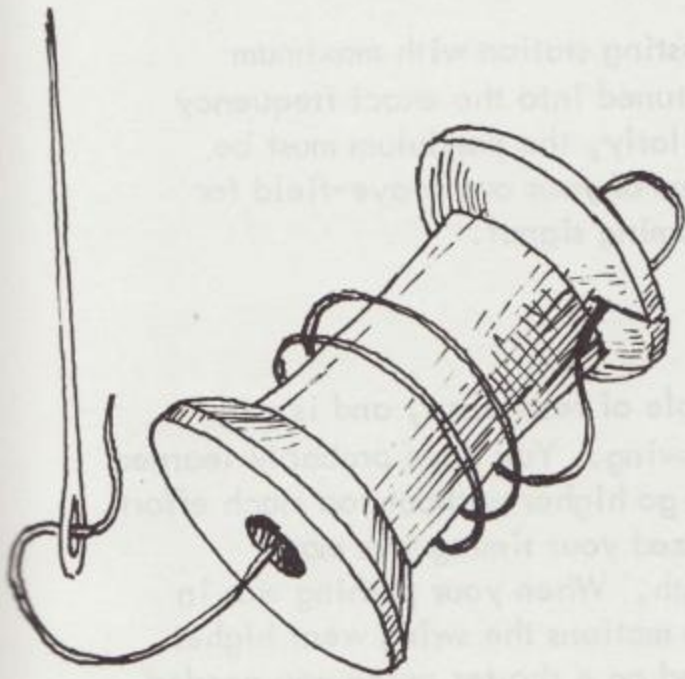
HOW DO I MAKE ONE?

Since every material has its own particular wave-field, it follows that the material of a pendulum can produce strong enough radiation to interfere with the wave-field under investigation. For this reason, keys or other metal objects are not recommended for use as pendulum weights, for their wave-fields are extremely strong, and erratic results occur when they are used. Wood is a more suitable material since it is inert.

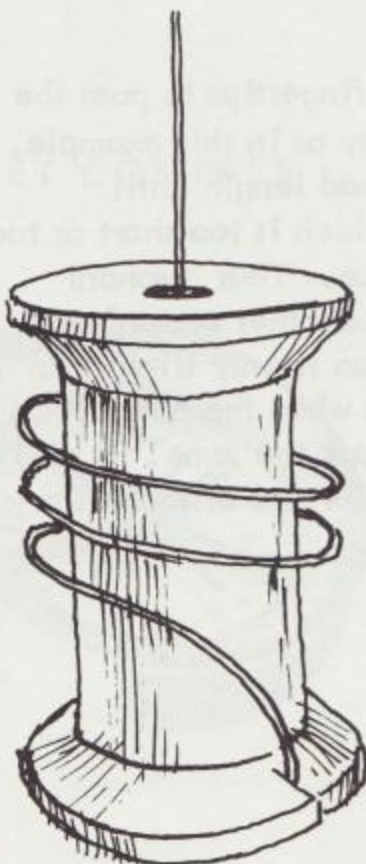
All you need to construct this pendulum is a new spool of black or green thread and a needle. A small spool is best. The one I use is about 1 1/4" long and 3/4" in diameter. The spool must be wooden and not styrofoam or plastic. The wooden part of the spool should be black. The reason for these particular colors will become apparent in later instructions.

Unwind about one foot of thread from the spool and attach it in the notch at the edge of the spool to prevent further uncoiling. Next pass the thread end through a needle and push it through the center of the spool. In doing so, keep the holes made through the paper in the spool ends as small as possible for pendulum balance. Remove the needle and suspend the spool by its thread. You have now made an efficient pendulum for yourself.

If you do not wish to make a pendulum, many models are available from the University of the Trees Press.



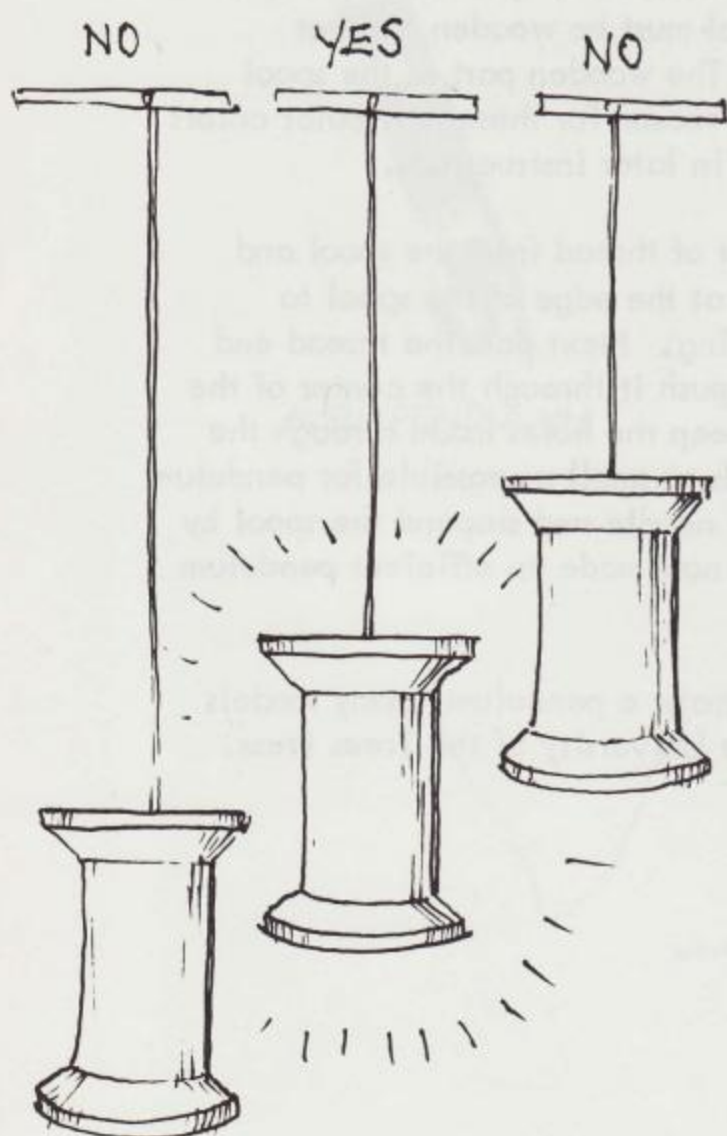
THE SPOOL PENDULUM



WHY IS TUNING IMPORTANT?

To receive a broadcasting station with maximum clarity, the radio is tuned into the exact frequency of that station. Similarly, the pendulum must be tuned to the vibrations of your own wave-field for it to amplify the incoming signal.

THE IMPORTANCE OF RESONANCE



This is another example of resonance, and is very similar to pushing a swing. You have probably learned how to make a swing go higher without too much effort. In doing so you realized your timing was more important than strength. When your pushing was in tune with the swing's motions the swing went higher. You may have noticed on a shorter swing you needed to push more often. This shows that the length of the swing is an important factor in resonance.

We need the vibrations in our fingertips to push the pendulum in much the same way as in this example. To do this, we change the thread length until resonance occurs. A length which is too short or too long will not produce resonance. Your resonant length will not be the same as another person's, since our vibrations differ, even if only slightly in some cases. Resonance occurs when the natural vibration rates of two objects are the same. It also occurs when the vibration rate of one of them is a multiple of the source.

TEST AREA

Before you begin to tune your pendulum a test area must be prepared. This work must be done in a room with a minimum of metal, electrical, and highly colored objects, vis. typewriters, radios, file drawers, clocks, refrigerators, and chemicals, which emit such strong radiations that the interference from these sources can cause erratic results for the student with limited experience. Among the brightly colored objects that are often present are curtains, books, and carpets.

Spectral green is a neutral color and does not cause interference. Placing a neutral green cloth across your table actually helps mask out undesirable waves. Open slightly any drawers of the table, since closed cavities affect the wave-field you are about to detect. Make sure the green is not too yellow or too blue but in between. This is important otherwise use black cloth or plain white. These two latter colors, however, condition and interact with the environment in a complementary way so your results would need cross checking when using black or white cloths.

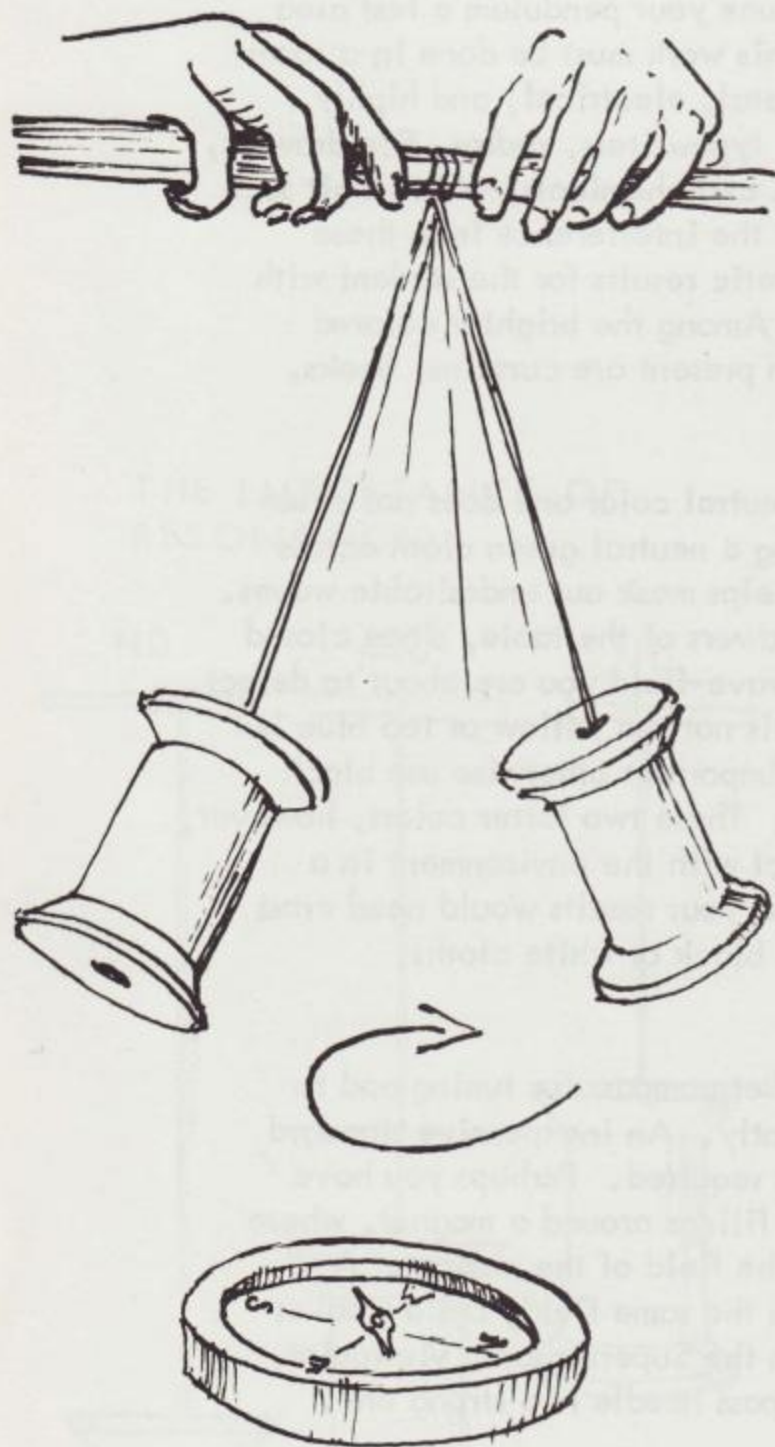
POCKET COMPASS



You will need a pocket compass for tuning and to practice with constantly. An inexpensive standard compass is all that is required. Perhaps you have seen pictures of iron filings around a magnet, where the filings outlined the field of the magnet. A compass needle emits the same field, but a weaker one. However, from the Supersensonic viewpoint, the field from a compass needle is a strong one.

To tune the pendulum, we hold it over the compass and change the thread length. When the pendulum swings spontaneously speed up, you have reached the resonant length.

THE TUNING PROCEDURE



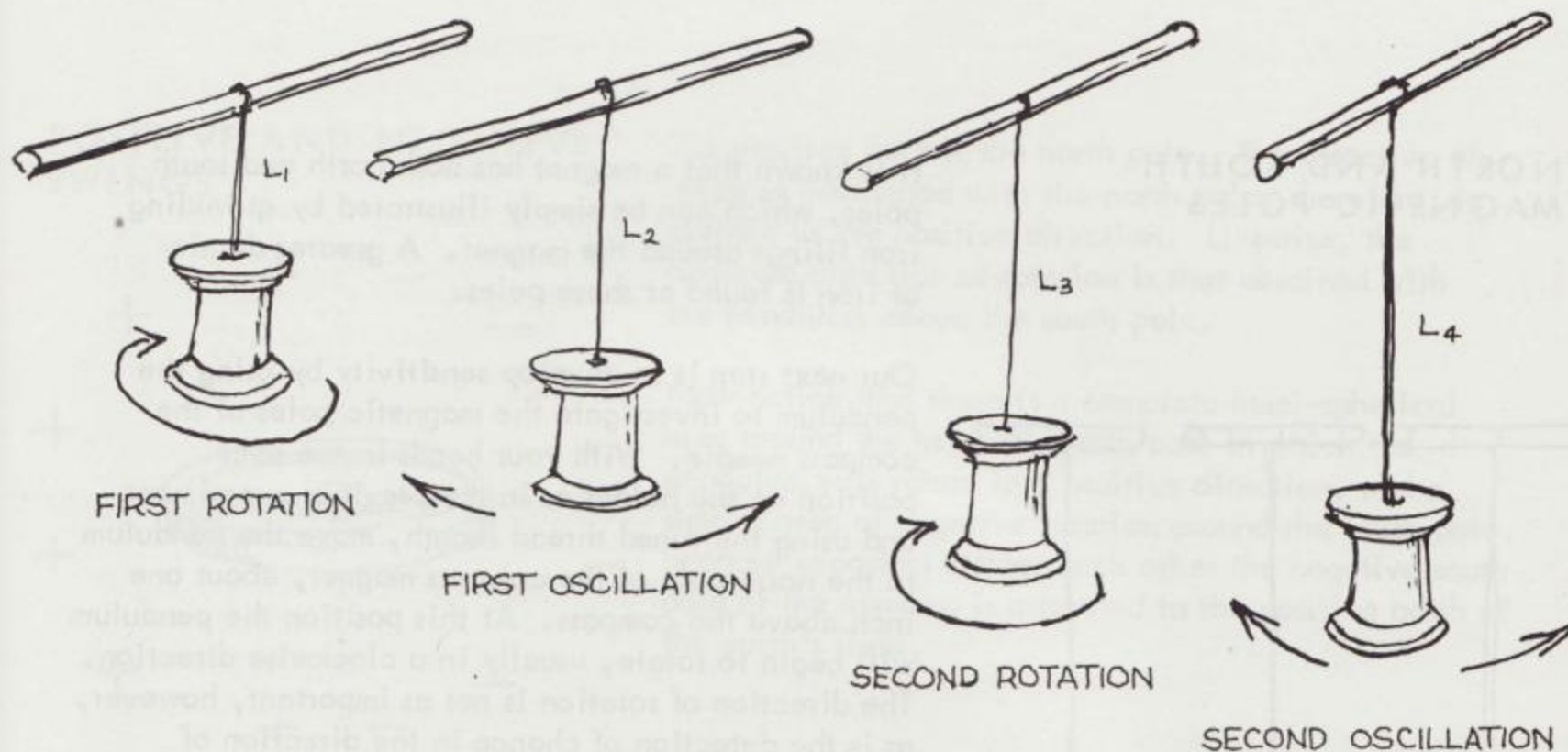
Wind the pendulum thread onto the middle of a thin unpainted stick of about $1/8$ inch cross-section and six inches long. A large firelighter matchstick without its tip is ideal for this purpose. Place the compass on a bare table or the floor and without resting your elbows on the table grip the holder lightly between the thumb and fingers of both hands, as illustrated, and hang the pendulum over the compass just east or west of its center, and about one inch above the compass. To aid this, align the north of the compass dial with the north pole of the compass needle.

Your best results will be obtained when you are facing one of the north, south, east or west coordinates. Remembering it is your body's nervous system that is detecting this radiation, it is natural that its orientation to the earth's magnetic field will affect the results. Find the position which most suits you and stay with it.

Give the pendulum a slight to and fro (oscillation) movement in the east-west direction. Relax your hand and arm muscles and gradually unwind the thread, raising the holder and hands to maintain the one-inch distance between the pendulum and compass. You will reach a point at which the pendulum starts to rotate in a circular motion. It will continue to do so as you lengthen the thread. This is the first rotation. As you unwind the pendulum slightly, it will begin to oscillate back and forth. This is the first oscillation.

Further lengthening the thread will cause the pendulum to perform a second rotation.

The unwinding is continued slightly until a second oscillation position is found. The length of thread at which this second oscillation occurs is the correct length for tuning.



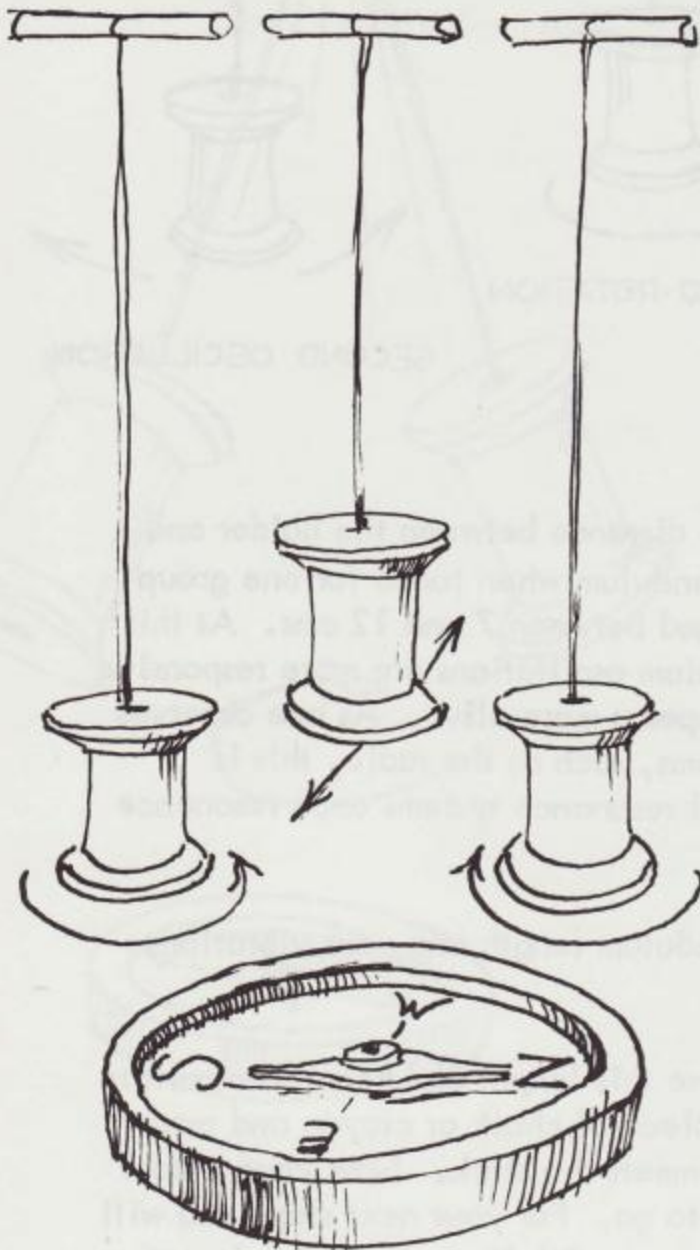
As an example, the distance between the holder and the center of the pendulum when tuned for one group of students has ranged between 7 and 12 cms. At this position, the pendulum oscillations are more responsive -- the pendulum appears more alive. As one observes in other tuned systems, such as the radio, this is characteristic of all resonance systems once resonance is reached.

In this case the pendulum length and your vibrations are in resonance.

Take care not to lose this length and hence necessitate retuning. Take a piece of chalk or crayon and mark the string where it meets the stick. Now your system is tuned and ready to go. For your next step, you will learn to increase your sensitivity to the detection of wave-fields, starting with the wave-field from your compass magnet.

HOW DO I DEVELOP SENSITIVITY?

NORTH AND SOUTH MAGNETIC POLES



It is known that a magnet has both north and south poles, which can be simply illustrated by sprinkling iron filings around the magnet. A greater density of iron is found at these poles.

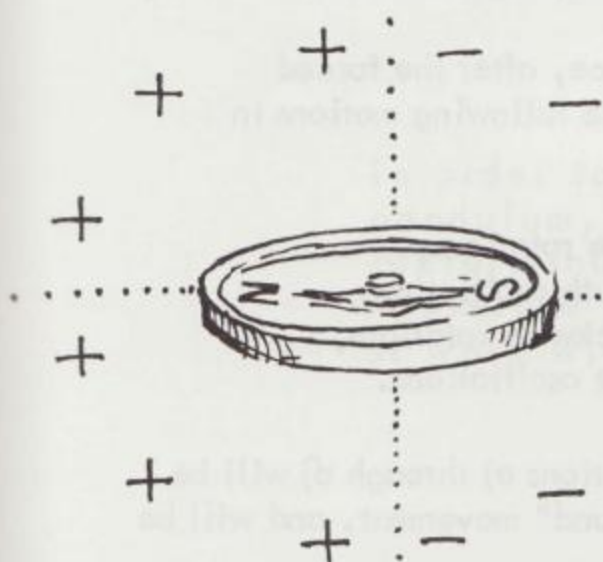
Our next step is to develop sensitivity by using the pendulum to investigate the magnetic poles of the compass needle. With your hands in the same position on the holder as in the previous experiment and using the tuned thread length, move the pendulum to the north pole of the compass magnet, about one inch above the compass. At this position the pendulum will begin to rotate, usually in a clockwise direction. The direction of rotation is not as important, however, as is the detection of change in the direction of rotation when the pendulum is moved to a position over the magnet's south pole.

Now practice moving the pendulum gently from the north pole to the south pole, observing the changes in the pendulum rotations. As you continue in this manner you will notice the pendulum changes from clockwise to anti-clockwise rotations as you pass it over the center of the magnet.

Repeat this motion but, in addition to momentary stops over the poles, also pause at the magnet's center and observe an east-west oscillation at this point.

When you are able to detect clockwise rotation, back and forth oscillation, and anti-clockwise rotation in that order, you have started to develop sensitivity to these wave-fields.

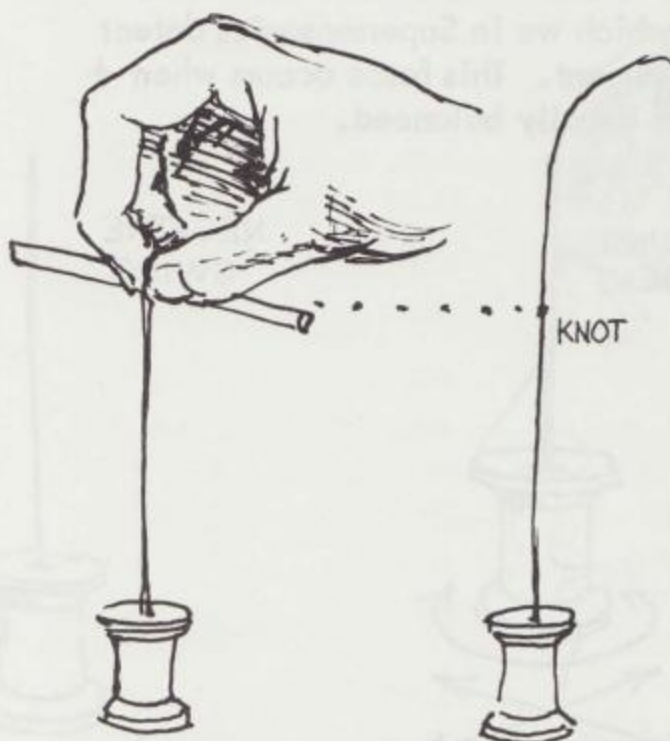
POSITIVE AND NEGATIVE SWINGS



The positive pole is the north pole. The direction of rotation associated with the north pole, therefore, is defined as the positive direction. Likewise, the negative direction of rotation is that obtained with the pendulum above the south pole.

Now notice that there is a complete hemi-spherical area around the magnet's north pole in which the pendulum will rotate in a positive direction, and a similar area of negative rotation around the south pole. Because opposites attract each other the negative south pole of the compass is attracted to the positive north of the earth's field.

REFINING THE TECHNIQUE



Touching the holder with as many fingers as possible brings a large number of nerve endings into play providing conduction of the signal to the pendulum. As sensitivity increases, we can eliminate the holder and the necessity of using both hands, as follows:

Grasp the holder at its center with the thumb and first finger of one hand and repeat the experiment until you can readily detect the three different effects over the magnet.

For the last step, tie a single knot in the thread to indicate your tuned length, and discarding the holder, grip the knot between the thumb and first finger. Drape the remainder of the thread over the back of your hand, as illustrated, to prevent it touching the active portion. To differentiate the three different components of the magnet's field in this way shows a degree of competency in wave-field detection. The time to master this lesson will vary from a few hours to a week, depending on whether you have an instructor or experiment by yourself.

COMPOUND MOVEMENT

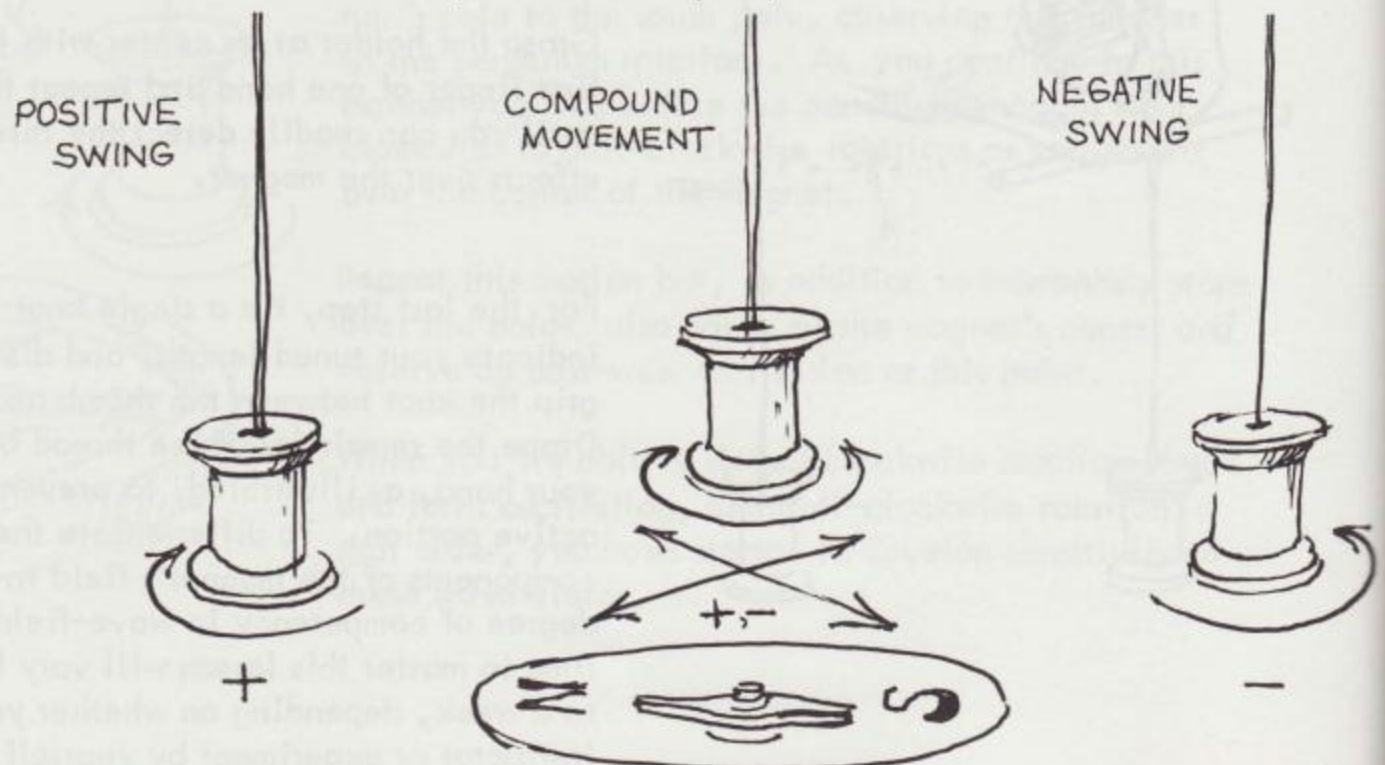
One more experiment to be undertaken is the investigation of the central portion of the magnet's wave-field. Place the pendulum over this point and set it oscillating gently so that the static friction of the pendulum and the air may be overcome.

The pendulum's performance, after the forced oscillations, consists of the following motions in sequence:

- a) clockwise rotations,
- b) north-south oscillations,
- c) anti-clockwise rotations,
- d) east-west oscillations.

The total of the above motions a) through d) will be referred to as the "compound" movement, and will be symbolized by $+$, $-$.

As we have demonstrated with the compass magnet, in nature there are $+$ and $-$ forces. Nature also has another force which we in Supersensonics detect as a compound movement. This force occurs when $+$ and $-$ portions are equally balanced.





YES



The basic yes/no chart...

